

Perfectoid Spaces and the Local Frobenius: Tilting, Prisms, and the Arithmetic Wall at p

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Abstract

Perfectoid spaces, introduced by Scholze [1], are the geometric framework that dissolves the Arithmetic Wall locally at a prime p . The core mechanism is *tilting*: every perfectoid field K of characteristic 0 has a characteristic- p twin $K^\flat$, and the étale topology of K and $K^\flat$ are identical. This provides a local Frobenius over $\mathbb{Q}_p$: the Frobenius of $K^\flat$ acts on the étale site shared with K . We build the full structure: perfectoid fields, the tilting functor, the almost purity theorem, perfectoid spaces, the Fargues–Fontaine curve, and prismatic cohomology (Bhatt–Scholze [5]). The Fargues–Fontaine curve is a geometric object whose Frobenius encodes the Weil group of $\mathbb{Q}_p$; prismatic cohomology is a universal p -adic cohomology theory specializing to de Rham, crystalline, and étale cohomology simultaneously. The key limitation is also identified: perfectoid geometry provides a Frobenius at each prime p separately but not a global Frobenius for $\mathrm{Spec} \mathbb{Z}$. The Arithmetic Wall of [7] is dissolved locally but not globally. Fargues–Scholze’s geometrization of local Langlands [4] is the current frontier: it proves the local Langlands correspondence for $\mathrm{GL}_n(\mathbb{Q}_p)$ via the geometry of the Fargues–Fontaine curve, but the passage from local to global remains the open step.

1 Perfectoid Fields

Definition 1 (Perfectoid field). A *perfectoid field* is a complete non-archimedean valued field K with value group $|K^\times| \subset \mathbb{R}_{>0}$ not discrete, satisfying:

1. $\mathrm{char}(K) = 0$ and $|p| < 1$ (or $\mathrm{char}(K) = p$), and
2. the Frobenius map $\varphi : \mathcal{O}_K/p \rightarrow \mathcal{O}_K/p$, $x \mapsto x^p$, is surjective.

Examples: $\mathbb{Q}_p(p^{1/p^\infty})^\wedge$, $\mathbb{C}_p$, $\mathbb{F}_p((t^{1/p^\infty}))^\wedge$.

Remark 2 (Why Frobenius surjectivity matters). In a perfectoid field, every element of $\mathcal{O}_K/p$ has a p -th root. This means the Frobenius $x \mapsto x^p$ is not merely injective (as in any characteristic- p ring) but also surjective: the field is “as close to characteristic p as a characteristic-0 field can be.” This is the key: we are in characteristic 0 but Frobenius acts.

Definition 3 (Tilting). Let K be a perfectoid field of characteristic 0. Its *tilt* is

$$K^\flat = \varprojlim_{x \mapsto x^p} K = \left\{ (x^{(0)}, x^{(1)}, x^{(2)}, \dots) : x^{(i)} \in K, (x^{(i+1)})^p = x^{(i)} \right\}.$$

The tilt $K^\flat$ is a perfectoid field of characteristic p . There is a canonical multiplicative (not additive) map $K^\flat \rightarrow K$, $(x^{(n)}) \mapsto x^{(0)}$.

Theorem 4 (Fontaine–Wintenberger, Scholze [1]). *Let K be a perfectoid field of characteristic 0 and $K^\flat$ its tilt. Then there is a canonical equivalence of categories:*

$$\text{Ét}(K) \xrightarrow{\sim} \text{Ét}(K^\flat),$$

where $\text{Ét}(F)$ is the category of finite étale F -algebras. In particular, $G_K \cong G_{K^\flat}$.

Remark 5 (Frobenius as a consequence of tilting). The tilt $K^\flat$ has characteristic p , so the absolute Frobenius $\varphi : x \mapsto x^p$ is a well-defined automorphism of $K^\flat$. Via the tilting equivalence (Theorem 4), φ acts on $G_K = G_{K^\flat}$. This is the local Frobenius over K : it lives in characteristic p but governs the étale topology of the characteristic-0 field K . For $K = \mathbb{Q}_p(p^{1/p^\infty})^\wedge$, $K^\flat = \mathbb{F}_p((p^{1/p^\infty}))^\wedge$, and the Frobenius of $K^\flat$ is the p -power map on the tilt — the characteristic- p avatar of the p -adic Frobenius.

2 Almost Purity

Theorem 6 (Almost purity theorem, Faltings–Scholze [1]). *Let R be a perfectoid K -algebra (a complete topological K -algebra whose underlying field is perfectoid in a suitable sense). Then the tilting functor*

$$S \mapsto S^\flat = \varprojlim_{x \mapsto x^p} S$$

induces an equivalence between finite étale R -algebras and finite étale $R^\flat$ -algebras. Moreover, the discriminant of any finite étale R -algebra is almost a unit: it differs from 1 only by elements of $\mathfrak{m}_K = \{x \in \mathcal{O}_K : |x| < 1\}$.

Remark 7 (Faltings’ original almost purity). Faltings [6] proved a weaker form: for the ring of integers of a perfectoid field, finite étale extensions have “almost” no ramification. Scholze’s version extends this to all perfectoid algebras and is the engine behind the primitive comparison theorem of p -adic Hodge theory.

3 Perfectoid Spaces

Definition 8 (Perfectoid algebra). A Banach $\mathbb{Q}_p$ -algebra R is *perfectoid* if $\mathcal{O}_R = \{x \in R : |x| \leq 1\}$ is $\mathfrak{m}$ -adically complete for $\mathfrak{m} = \{x : |x| < 1\}$, and the Frobenius $\varphi : \mathcal{O}_R/p \rightarrow \mathcal{O}_R/p$ is surjective.

Definition 9 (Perfectoid space). A *perfectoid space* is an adic space X (a locally ringed topological space modeled on $\text{Spa}(R, R^+)$ for Huber pairs (R, R^+)) that is locally isomorphic to $\text{Spa}(R, R^+)$ with R perfectoid. The *tilt* $X^\flat$ is the perfectoid space of characteristic p obtained by tilting each local perfectoid algebra.

Theorem 10 (Tilting equivalence for spaces [1]). *For a perfectoid space X over $\mathbb{Q}_p$, there is a homeomorphism $|X| \cong |X^\flat|$ of underlying topological spaces and an equivalence $X_{\text{ét}} \simeq X_{\text{ét}}^\flat$ of small étale sites.*

Theorem 11 (Primitive comparison theorem [2]). *Let X be a proper smooth rigid-analytic variety over $\mathbb{C}_p$. There is a canonical, $G_{\mathbb{Q}_p}$ -equivariant isomorphism*

$$H_{\text{ét}}^i(X_{\mathbb{C}_p}, \mathbb{Z}_p) \otimes_{\mathbb{Z}_p} B_{\text{dR}} \cong H_{\text{dR}}^i(X) \otimes_{\mathbb{Q}_p} B_{\text{dR}},$$

recovering Fontaine’s de Rham comparison conjecture and extending it to the rigid-analytic setting via perfectoid methods.

4 The Fargues–Fontaine Curve

Definition 12 (Fargues–Fontaine curve [3]). Let E be a finite extension of $\mathbb{Q}_p$ and $F = E^\flat$ its tilt (a local field of characteristic p). The *Fargues–Fontaine curve* is the scheme

$$\text{FF}_E = \left(\text{Spec } B_{\text{crys}}^{\varphi=1} \setminus \{0\} \right) / \varphi^{\mathbb{Z}},$$

where B_{crys} is Fontaine’s crystalline period ring attached to F and φ is the Frobenius. Geometrically, FF_E is a complete curve over E (in the sense of algebraic geometry) with one distinguished closed point ∞ corresponding to the de Rham period ring B_{dR} .

Theorem 13 (Structure of FF_E [3]). *1. FF_E is a regular Noetherian scheme of dimension 1.*

2. Vector bundles on FF_E are classified by Harder–Narasimhan slopes; the category of semistable bundles of slope 0 is equivalent to the category of crystalline representations of G_E .

3. The Weil group W_E acts on FF_E via the Frobenius of F . The Frobenius φ_E of FF_E encodes the arithmetic of E .

Remark 14 (The curve as local Frobenius machine). The Fargues–Fontaine curve is the geometric object whose Frobenius replaces the missing Frobenius of p -adic Hodge theory. On FF_E , the Frobenius φ acts as a genuine geometric automorphism. The crystalline period ring B_{crys} is the ring of functions on $\text{FF}_E \setminus \{\infty\}$; the de Rham ring B_{dR} is the completed local ring at ∞ . Every Fontaine period ring is a piece of the geometry of one curve.

5 Prismatic Cohomology

Definition 15 (Prism [5]). A *prism* is a pair (A, I) where A is a commutative ring equipped with a Frobenius lift $\varphi : A \rightarrow A$ (a ring map with $\varphi(x) \equiv x^p \pmod{p}$) and $I \subset A$ a principal ideal such that A is derived (p, I) -complete and $I + \varphi(I) \cdot A = A$ (the Frobenius is “invertible at I ”).

Theorem 16 (Prismatic cohomology [5]). *For a smooth p -adic formal scheme $\mathfrak{X}$ over $\mathcal{O}_K$ (K a p -adic field), there exists a cohomology theory $R\Gamma_\Delta(\mathfrak{X})$ valued in the derived category of $\mathbb{Z}_p$ -modules, equipped with a Frobenius φ , that simultaneously specializes to:*

<i>Specialization</i>	<i>Via</i>	<i>Cohomology</i>
$I = p$	$A/(p)$	Crystalline cohomology $H_{\text{crys}}^*(\mathfrak{X}_k)$
$I = \ker(\theta)$	$A/I \cong \mathcal{O}_K$	de Rham cohomology $H_{\text{dR}}^*(X_K)$
Invert I	$A[I^{-1}]$	Étale cohomology $H_{\text{ét}}^*(X_{\bar{K}}, \mathbb{Z}_p)$

The comparison isomorphisms between crystalline, de Rham, and étale cohomology all arise as specializations of the single prismatic object.

Remark 17 (Prism as universal Frobenius). The prismatic Frobenius $\varphi : R\Gamma_\Delta(\mathfrak{X}) \rightarrow R\Gamma_\Delta(\mathfrak{X})$ is the universal Frobenius for p -adic geometry. It is a genuine, canonical operator — not a conjecture, not a formal symbol — that governs all p -adic cohomology theories at once. This is the closest existing object to a “Frobenius for $\text{Spec } \mathbb{Z}_p$ ”: a single Frobenius acting on a universal cohomology theory for smooth p -adic schemes.

6 Fargues–Scholze: Geometrization of Local Langlands

Theorem 18 (Fargues–Scholze 2021 [4]). *Let G be a reductive group over $\mathbb{Q}_p$. There is a spectral action of the category of perfect complexes on the stack of L -parameters $[\text{Loc}_{\hat{G}}(\text{FF}_{\mathbb{Q}_p})]$ on the derived category $D(G(\mathbb{Q}_p), \mathbb{Z}_\ell)$ of smooth representations of $G(\mathbb{Q}_p)$. For $G = \text{GL}_n$, this realizes the local Langlands correspondence: smooth irreducible representations of $\text{GL}_n(\mathbb{Q}_p)$ correspond to n -dimensional ℓ -adic representations of the Weil group $W_{\mathbb{Q}_p}$.*

Remark 19 (Why Fargues–Scholze matters for the Arithmetic Wall). The Fargues–Scholze theorem proves the local Langlands correspondence using only the geometry of the Fargues–Fontaine curve $\text{FF}_{\mathbb{Q}_p}$. The Frobenius of the curve encodes the Weil group; the L -parameters are vector bundles on the curve. This is a genuine geometric proof of a statement (local Langlands) that previously required deep analytic methods (Henniart, Harris–Taylor). The local Arithmetic Wall — the absence of Frobenius at p — is dissolved by the curve.

7 What Remains: The Local-to-Global Gap

Proposition 20 (Perfectoid geometry does not cross the global Wall). *Perfectoid spaces, the Fargues–Fontaine curve, and prismatic cohomology provide a Frobenius for $\text{Spec } \mathbb{Z}_p$ but not for $\text{Spec } \mathbb{Z}$.*

1. *At each prime p : the prismatic Frobenius is a canonical operator on $H_\Delta^*(\mathfrak{X})$. The Fargues–Fontaine curve encodes the Weil group of $\mathbb{Q}_p$. Local Langlands is proved.*
2. *Globally: there is no Fargues–Fontaine curve for $\mathbb{Q}$. The product of local curves does not assemble into a global curve (this would be an “arithmetic surface” for $\text{Spec } \mathbb{Z}$, whose existence is the content of the global Langlands program).*

3. *The Arithmetic Wall [7] is the failure to assemble local Frobenii into a global one. BSD, RH, and the Tate conjecture all require crossing from local to global.*

Theorem 21 (Summary: perfectoid position in the hierarchy).

<i>Object</i>	<i>Scope</i>	<i>What it provides</i>
<i>Tilting equivalence</i>	<i>Local (p-adic fields)</i>	<i>Frobenius on étale site</i>
<i>Almost purity</i>	<i>Local (perfectoid algebras)</i>	<i>Unramified extensions</i>
<i>Primitive comparison</i>	<i>Local ($\mathbb{C}_p$)</i>	<i>B_{dR} comparison</i>
<i>Fargues–Fontaine curve</i>	<i>Local ($\mathbb{Q}_p$)</i>	<i>Geometric local Langlands</i>
<i>Prismatic cohomology</i>	<i>Local ($\mathbb{Z}_p$-schemes)</i>	<i>Universal p-adic cohomology</i>
<i>Global Langlands</i>	<i>Global ($\mathbb{Q}$)</i>	<i>Open</i>
<i>Bloch–Kato [8]</i>	<i>Global ($\mathbb{Q}$)</i>	<i>Open</i>
<i>BSD, RH, Tate</i>	<i>Global ($\mathbb{Q}$)</i>	<i>Open</i>

The frontier is the vertical line between local and global. Perfectoid geometry has fully colonized the left column. The right column is the Arithmetic Wall.

References

- [1] Scholze, P. (2012). Perfectoid spaces. *Publ. Math. IHES* 116, 245–313.
- [2] Scholze, P. (2013). p -adic Hodge theory for rigid-analytic varieties. *Forum Math. Pi* 1, e1.
- [3] Fargues, L., Fontaine, J.-M. (2021). *Courbes et fibrés vectoriels en théorie de Hodge p -adique*. Astérisque 406.
- [4] Fargues, L., Scholze, P. (2021). Geometrization of the local Langlands correspondence. Preprint, arXiv:2102.13459.
- [5] Bhatt, B., Scholze, P. (2022). Prisms and prismatic cohomology. *Ann. Math.* 196, 1135–1275.
- [6] Faltings, G. (2002). Almost étale extensions. *Astérisque* 279, 185–270.
- [7] Rincón, D. (2026). The Arithmetic Wall: Why the Millennium Problems Resisted Solution, and What the Resistance Reveals. Preprint.
- [8] Rincón, D. (2026). The Bloch–Kato Conjecture: Tamagawa Numbers, Selmer Groups, and Motives. Preprint.
- [9] Rincón, D. (2026). The Langlands Thread. Preprint.